AKASH S

+91 8925655820 akashs231005@gmail.com Akash231005 Akash—S
akash-s-836200333 Chennai, India

SUMMARY

Computer Science undergraduate with strong foundations in Data Structures, OOP, DBMS, Operating Systems, Computer Networks, and Machine Learning. Experienced in Java, Python, JavaScript, React, Node.js, SQL, REST APIs, and full-stack development. Hands-on experience building AI/ML solutions using Scikit-learn, TensorFlow, Keras, YOLOv8, and OpenCV across computer vision, object detection, image classification, regression, and time-series forecasting. Experienced in automated testing, cloud deployment, Git, Docker, and CI/CD.

TECHNICAL SKILLS

Programming Languages: Java, Python, JavaScript, C
Core CS: Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks
AI/ML: Scikit-learn, TensorFlow, Keras, YOLOv8, OpenCV, NumPy, Pandas, Joblib
Web & Backend: React, Node.js, Express.js, Spring Boot, REST APIs, HTML5, CSS3, Flask
Databases: Oracle SQL, MySQL, MongoDB, SQLite
Testing: Playwright, Jest, Supertest, Newman, Postman
DevOps & Cloud: Docker, GitHub Actions, Vercel, Render
Developer Tools: Git, GitHub, VS Code

PROJECTS

AgriFleet – Smart Agricultural Fleet Management Platform (React, Node.js, Express.js, MongoDB, JWT) GitHub Repository / Live Demo

- Built a full-stack MERN platform for managing agricultural tractor services, enabling farmer bookings, driver assignment, fleet tracking, and end-to-end job management.
- Implemented JWT-based role-based access control, smart tractor-driver allocation, fleet maintenance tracking, operational dashboards, and secure REST APIs with React, Redux Toolkit, Node.js, Express.js, and MongoDB.

Placement Progress Tracker – Full-Stack MERN Application (React, Node.js, Express.js, MongoDB, Playwright) GitHub Repository / Live Demo

- Developed a preparation tracking platform for DSA, core CSE, aptitude, typing, mock interviews, and job applications with analytics-driven progress dashboards.
- Built secure REST APIs with JWT authentication and bcrypt, automated backend/frontend testing using Jest, Supertest, and Playwright, and implemented CI/CD with GitHub Actions and Vercel/Render deployment.

Tesla Autopilot Clone – Road Object Detection (Python, YOLOv8, PyTorch, OpenCV, KITTI) GitHub Repository

- Built a YOLOv8-based autonomous-driving perception pipeline on the KITTI dataset to detect cars, pedestrians, cyclists, trucks, vans, trams, and other road objects.
- Automated KITTI-to-YOLO annotation conversion, bounding-box validation, and dataset splitting; fine-tuned YOLOv8 using transfer learning and evaluated detection using Precision, Recall, and mAP.

Face Recognition System using Deep Face Embeddings (Python, TensorFlow, MTCNN, EfficientNet-Lite, SVM) GitHub Repository

- Developed an LFW-based face recognition pipeline using MTCNN for face detection and EfficientNet-Lite for deep facial feature extraction.
- Trained a Linear SVM classifier on extracted embeddings and implemented preprocessing, model persistence, inference, and evaluation using accuracy, classification reports, and confusion matrices.

EDUCATION

2024 – Present **B.E. Computer Science and Engineering**, SSN College of Engineering, Kancheepuram
CGPA: **7.995** **Academic Status: No Active Backlogs**

2022 – 2023 **Higher Secondary Certificate (HSC)**, Rajavignesh Hr Sec School, Melamathur Percentage: **96.67%**

2020 – 2021 **Secondary School Leaving Certificate (SSLC)**, Rajavighesh Hr Sec School, Melamathur All Pass (Half-yearly: **98.0%**)

CERTIFICATIONS

Artificial Intelligence and Machine Learning – GUVI — HCL **Verify Certificate**

Full-Stack Web Development – Jyesta — E-Cell, IIT Roorkee **Verify Certificate**